

Monthly Stipend Tracker for Scholarship Students - Project Report

1.0 Executive Summary

This report documents the development and functionality of the Monthly Stipend Tracker, a command-line application designed to help scholarship students monitor and manage their monthly stipend allocations. The program allows students to track expenses, calculate balances, analyze spending patterns, and maintain historical records of their financial management.

2.0 Project Information

Detail Description
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Project Title Monthly Stipend Tracker for Scholarship Students
Course CCC1123 - Problem Solving and Programming Fundamental
Project Type Mini Programming Project
Development Date January 2026
Programming Language C Programming Language

3.0 Problem Statement

Scholarship students receiving monthly stipends often struggle with:

- Lack of systematic tracking of stipend utilization
- Difficulty in monitoring food vs. essential expenses
- No historical record of monthly spending patterns
- Inability to analyze percentage allocation of funds
- Manual calculation errors in determining remaining balances

4.0 Program Features and Functionality

4.1 Core Features

- Monthly Record Management: Add and store up to 12 months of stipend data
- Automated Calculations: Automatic calculation of totals, balances, and percentages
- Data Persistence: File-based storage for long-term data retention
- Comprehensive Reporting: Multiple view options for data analysis
- Budget Monitoring: Warnings for overspending and low balances

4.2 Menu Options

Option	Function	Description
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1	Add Monthly Record	Input monthly stipend and expense data
2	View Expense Summary	Tabular display of all monthly records
3	View Monthly Report	Detailed report for a specific month
4	Save Records	Manual save to text file
5	View Statistics	Overall spending statistics
6	Exit Program	Exit with auto-save functionality

5.0 Technical Implementation

5.1 Data Structure

```
typedef struct {
    char month[20];           /* Month name */
    float stipend;             /* Monthly stipend amount */
    float food_expenses;       /* Food expenses */
    float other_expenses;      /* Other essential expenses */
    float total_expenses;      /* Calculated total expenses */
    float balance;             /* Remaining balance */
    float food_percentage;     /* Percentage spent on food */
    float other_percentage;    /* Percentage spent on other expenses */
} StipendRecord;
```

5.2 Key Algorithms

1. Expense Calculation Algorithm:

- Total Expenses = Food Expenses + Other Expenses
- Balance = Stipend - Total Expenses
- Food Percentage = (Food Expenses / Total Expenses) × 100
- Other Percentage = (Other Expenses / Total Expenses) × 100

2. File I/O Operations:

- Sequential file storage in `stipend\_records.txt`

- Automatic loading on program startup
- Auto-save on program exit

5.3 Functions Overview

Function	Purpose	Parameters
displayMenu()	Display main menu interface	None
inputData()	Collect monthly data from user	Records array, count pointer
calculateExpenses()	Perform all financial calculations	Pointer to StipendRecord
displaySummary()	Show tabular summary of all months	Records array, count
saveToFile()	Save data to external file	Records array, count
loadFromFile()	Load data from external file	Records array, count pointer
displayMonthlyReport()	Show detailed monthly analysis	Records array, count
calculateTotalStipend()	Calculate cumulative stipend	Records array, count
calculateAverageExpenses()	Compute average monthly spending	Records array, count

6.0 User Interface Design

6.1 Visual Elements

- Border Design: Uses ASCII characters for clean borders
- Formatting: Tabular displays with proper alignment
- Color Indicators: Text-based indicators ( ,  , )
- Section Separation: Clear visual separation between sections

6.2 Input Validation

- Maximum record limit (12 months)
- File operation error handling

- Invalid menu choice detection
- Month selection validation

7.0 Sample Program Output:

```

||
||  MONTHLY STIPEND TRACKER FOR SCHOLARSHIP STUDENTS  ||
||
  
```

MAIN MENU

1.  Add Monthly Stipend Record
 2.  View Expense Summary
 3.  View Monthly Report
 4.  Save Records to File
 5.  View Statistics
 6.  Exit Program
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Expense Summary Table:

EXPENSE SUMMARY							
Month	Stipend	Food	Other	Total	Balance	Food%	Other%
January	400.00	200.00	200.00	400.00	0.00	50.0%	50.0%
Feburay	450.00	200.00	150.00	350.00	100.00	57.1%	42.9%
March	450.00	200.00	150.00	350.00	100.00	57.1%	42.9%
April	400.00	200.00	100.00	300.00	100.00	66.7%	33.3%
May	450.00	200.00	150.00	350.00	100.00	57.1%	42.9%
June	450.00	200.00	0.00	200.00	250.00	100.0%	0.0%
July	450.00	200.00	200.00	400.00	50.00	50.0%	50.0%
August	450.00	200.00	150.00	350.00	100.00	57.1%	42.9%
September	400.00	150.00	200.00	350.00	50.00	42.9%	57.1%
October	400.00	200.00	100.00	300.00	100.00	66.7%	33.3%
November	400.00	200.00	200.00	400.00	0.00	50.0%	50.0%
December	450.00	200.00	200.00	400.00	50.00	50.0%	50.0%

Monthly Expenses Report:(if you spend less than overall stipend)

MONTHLY REPORT FOR Feburary		
Monthly Stipend:	RM	400.00
Food Expenses:	RM	200.00 (50.0%)
Other Expenses:	RM	200.00 (50.0%)
Total Expenses:	RM	400.00
Remaining Balance:	RM	0.00
 NOTE: You have less than 20% of stipend remaining.		

Monthly Expenses Report(IF you send over the stipend)

MONTHLY REPORT FOR April		
Monthly Stipend:	RM	400.00
Food Expenses:	RM	250.00 (50.0%)
Other Expenses:	RM	250.00 (50.0%)
Total Expenses:	RM	500.00
Remaining Balance:	RM	-100.00
 WARNING: You have overspent by RM 100.00 this month!		

Overall Statistics:

```
Enter your choice: 5

📊 Overall Statistics:
=====
Total Months Tracked: 4
Total Stipend Received: RM 1650.00
Average Monthly Expenses: RM 412.50
```

8.0 Error Handling and Validation

8.1 Implemented Validations

- Maximum record capacity check
- File read/write error handling
- Invalid menu option handling
- Month selection range validation
- Division by zero prevention in percentage calculations

8.2 User Feedback Messages

- Success confirmations with 
- Warning messages with 
- Error notifications with 
- Informational messages with , ,  icons

9.0 File Storage Format

Data Structure in `stipend\_records.txt`:

January 1500.00 600.00 400.00 1000.00 500.00 60.00 40.00

February 1500.00 700.00 500.00 1200.00 300.00 58.33 41.67

Field Order: Month, Stipend, Food, Other, Total, Balance, Food%, Other%

10.0 Limitations and Assumptions

10.1 Current Limitations

- Fixed maximum of 12 monthly records
- No student identification system
- No data editing or deletion functionality
- Single expense category for "other" expenses
- No graphical user interface

10.2 Design Assumptions

- Monthly stipend is fixed per month

- Expenses categorized only as food or other essentials
- All amounts in Malaysian Ringgit (RM)
- Data is numeric and non-negative

11.0 Potential Enhancements

11.1 Short-term Improvements

1. Add record editing and deletion capabilities
2. Implement search functionality by month
3. Add data validation for negative amounts
4. Include monthly budget planning feature

11.2 Long-term Enhancements

1. Multi-user support with login system
2. Graphical visualization of spending patterns
3. Export to spreadsheet format (CSV)
4. Mobile application version
5. Cloud-based data synchronization

12.0 Learning Outcomes

This project demonstrates proficiency in:

- Structured Programming: Modular function design
- Data Structures: Effective use of structures and arrays
- File Handling: Persistent data storage and retrieval
- User Interface: Creating user-friendly command-line interfaces
- Problem Solving: Translating real-world problems into algorithmic solutions

13.0 Conclusion

The Monthly Stipend Tracker successfully addresses the financial management needs of scholarship students through a well-structured, functional command-line application. The program combines practical utility with solid programming principles, providing a foundation that could be expanded into a more comprehensive financial management tool. Its modular design allows for easy maintenance and future enhancements, making it a valuable learning project in problem-solving and programming fundamentals.